

Roll No: 2303A52189

## SQL JOIN Practice – Tables and 10 Questions

Use the following tables for all questions.

### 1. departments

| dept_id | dept_name   | location  |
|---------|-------------|-----------|
| 10      | Engineering | Chennai   |
| 20      | Sales       | Hyderabad |
| 30      | HR          | Bengaluru |
| 40      | Research    | Pune      |
| 50      | Finance     | Mumbai    |

### 2. employees

| emp_id | emp_name | dept_id | manager_id | salary | hire_date  |
|--------|----------|---------|------------|--------|------------|
| 101    | Arjun    | 10      | NULL       | 90000  | 2021-01-15 |
| 102    | Priya    | 10      | 101        | 65000  | 2022-03-10 |
| 103    | Karthik  | 20      | 106        | 55000  | 2022-07-20 |
| 104    | Meena    | 30      | NULL       | 60000  | 2020-11-05 |
| 105    | Rahul    | NULL    | 101        | 50000  | 2023-02-12 |
| 106    | Divya    | 20      | NULL       | 85000  | 2019-06-25 |
| 107    | Sanjay   | 10      | 101        | 72000  | 2021-09-17 |
| 108    | Anitha   | 40      | NULL       | 78000  | 2020-04-30 |
| 109    | Vikram   | 20      | 106        | 48000  | 2024-01-08 |

### 3. projects

| project_id | project_name    | dept_id | budget |
|------------|-----------------|---------|--------|
| 201        | Banking App     | 10      | 500000 |
| 202        | Sales Portal    | 20      | 300000 |
| 203        | AI Research     | 40      | 800000 |
| 204        | Employee System | 30      | 250000 |
| 205        | Cloud Migration | 10      | 600000 |

206          Audit Automation      50      350000

#### 4. `employee_project`

| <code>emp_id</code> | <code>project_id</code> | <code>hours</code> |
|---------------------|-------------------------|--------------------|
| 101                 | 201                     | 120                |
| 102                 | 201                     | 100                |
| 102                 | 205                     | 80                 |
| 103                 | 202                     | 110                |
| 106                 | 202                     | 70                 |
| 107                 | 201                     | 90                 |
| 107                 | 205                     | 130                |
| 108                 | 203                     | 150                |
| 104                 | 204                     | 60                 |

## Questions

1. Write an `INNER JOIN` query to display every employee's ID, name, department name and department location. Employees without a department should not appear.
2. Write a `LEFT JOIN` query to display all employees along with their department names. Employees who are not assigned to a department must also be displayed.
3. Write a `LEFT JOIN` query to find departments that do not currently have any employees.
4. Write a query using multiple joins to display the employee name, project name, department name and number of hours worked.
5. Write a query to display all projects and the employees assigned to each project. Projects without assigned employees must also appear.
6. Using a self-join on the `employees` table, display each employee's name and their manager's name. Employees without managers must also appear.
7. Write a query to find employees who are working on more than one project. Display the employee ID, employee name and total number of projects.
8. Write a query to display each department's name, number of employees and average employee salary. Departments without employees must show an employee count of 0.
9. Write a query to find employees who work on projects belonging to a department different from their own department. Display the employee name, employee department, project name and project department.
10. Write a query to find projects that have no employees assigned. Display the project ID, project name, department name and budget.

Answers for the queries:

- 1)select e.emp\_id,e.emp\_name,d.dept\_name,d.location from employees e inner join departments d ON e.dept\_id=d.dept\_id;
- 2)select e.emp\_id,e.emp\_name,d.dept\_name from employees e left join departments d ON e.dept\_id=d.dept\_id;
- 3)select d.dept\_id,d.dept\_name from departments d left join employees e ON d.dept\_id=e.dept\_id where e.emp\_id IS NULL;
- 4)select e.emp\_name,p.project\_name,d.dept\_name,ep.hours from employees e join employee\_project ep ON e.emp\_id=ep.emp\_id join projects p ON ep.project\_id=p.project\_id JOIN departments d ON p.dept\_id=d.dept\_id;
- 5)select p.project\_id,p.project\_name,e.emp\_name from projects p left join employee\_project ep ON p.project\_id=ep.project\_id left join employees e ON ep.emp\_id=e.emp\_id;
- 6)select e.emp\_name,coalesce(m.emp\_name,'No manager') from employees e left join employees m ON e.manager\_id=m.emp\_id;
- 7)select e.emp\_id,e.emp\_name,count(ep.project\_id) as totalprojects from employees e join employee\_project ep ON e.emp\_id=ep.emp\_id GROUP by e.emp\_id,e.emp\_name HAVING Count(ep.project\_id)>1;
- 8)select d.dept\_name,count(e.emp\_id),coalesce(avg(e.salary),0) from departments d left join employees e ON d.dept\_id=e.dept\_id group by d.dept\_id,d.dept\_name;
- 10)select p.project\_id,p.project\_name,d.dept\_name,p.budget from projects p join departments d ON p.dept\_id=d.dept\_id left join employee\_project ep ON p.project\_id=ep.project\_id where ep.emp\_id is null;